Computer Vision

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Intrusion Detection System Using Change Detection Techniques

Project Report

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1 Introduction

1.1 Context and Motivation

Automatic intrusion detection represents a fundamental problem in computer vision with widespread applications in surveillance, home security, and environmental monitoring. The ability to automatically detect and classify moving objects in video streams enables intelligent monitoring systems that can operate continuously without human supervision, providing cost-effective solutions for security-critical scenarios.

Traditional surveillance systems rely on continuous human monitoring, which is expensive, prone to fatigue-induced errors, and impractical for large-scale deployments. Computer vision-based approaches address these limitations by enabling automated analysis of video streams, detecting anomalies, tracking objects of interest, and generating alerts when specific events occur.

This project develops a real-time intrusion detection system capable of identifying and classifying moving objects in indoor environments, with specific focus on distinguishing between persons and other objects (added or removed items). The approach is based on classical computer vision techniques—background subtraction, edge detection, and mathematical morphology—which, when properly combined, offer robustness and computational efficiency.

The system operates on a fixed-camera setup monitoring an indoor scene, making it particularly suitable for applications such as office monitoring, museum surveillance, or smart home security systems.

1.2 Project Objectives

The project addresses two main tasks with specific requirements:

1.2.1 Task 1: Change Detection (Mandatory)

The primary objective is to implement a comprehensive change detection system that produces both visual and textual outputs:

1. **Graphical Output:** Generate an annotated video showing detected objects with:
    - Precise contours or bounding boxes
    - Persistent tracking identifiers (IDs)
    - Classification labels (person vs. other)
2. **Textual Output:** Produce a structured CSV file containing per-frame information:
    - Frame index and timestamp
    - Number of detected objects

- Per-object features: ID, classification, geometric properties (area, perimeter, centroid, bounding box), shape descriptors (extent, solidity, aspect ratio)
- Tracking status (confirmed/tentative)

3. **Person/Object Classification:** Implement robust classification distinguishing persons from other objects based on:
   - Size constraints (minimum area and height)
   - Shape features (aspect ratio, solidity, extent)
   - Morphological characteristics
4. **Proper Change Detection:** Combine intensity-based and edge-based detection for robustness:
   - Background modeling with adaptive updating
   - Dual-threshold detection (intensity difference + edge difference)
   - Mathematical morphology for noise reduction
   - 8-connectivity labeling for component extraction

1.2.2 Task 2: Removed Object Detection (Optional)

Extend the system to detect objects removed from the scene:

- Implement inverse difference detection (background - current frame)
- Apply dedicated morphological filtering
- Maintain separate tracking for removed objects
- Provide visual distinction (different color coding) in output video
- Include removal flag in CSV output

1.2.3 Enhancement Objectives

Beyond the core requirements, the project aims to implement several improvements:

1. **Illumination Robustness:** CLAHE preprocessing to handle varying lighting conditions
2. **Optimized Morphology:** Elliptical kernels and multi-pass processing for cleaner results
3. **Robust Tracking:** IoU-based association with occlusion handling
4. **Contour Refinement:** Polygonal approximation for smoother visualization
5. **Analysis Logging:** Save intermediate processing steps for report generation

1.3 Document Organization

The remainder of this report is organized as follows:

- **Section 2:** Theoretical background covering change detection fundamentals, edge detection operators, mathematical morphology, and connected component labeling
- **Section 3:** Detailed description of the proposed system architecture, including background modeling, preprocessing pipeline, morphological filtering, classification strategy, and tracking approach, in the discussion are also presented the choice of the different parameters and implementation details. Pictures and plots are presented to enforce the performances obtained.
- **Section 4:** Summary of achieved objectives, key contributions, and future work directions

The report includes comprehensive visual documentation through nine figures generated from actual processing results, demonstrating the complete pipeline from raw input to final detection outputs.

2 Theoretical Background

This section provides the theoretical foundation for the techniques employed in the intrusion detection system. We cover the fundamental concepts of change detection, edge detection operators, mathematical morphology, and connected component analysis.

2.1 Change Detection Fundamentals

Change detection aims to identify significant differences between a reference frame (background model) and current observations, enabling the extraction of moving or newly appeared objects. This is a fundamental operation in computer vision with applications ranging from video surveillance to medical imaging.

2.1.1 Background Subtraction Approaches

Several strategies exist for modeling and updating the background:

Frame Differencing. The simplest approach computes the absolute difference between consecutive frames:

$$D_t(x, y) = |I_t(x, y) - I_{t-1}(x, y)| \tag{1}$$

While computationally efficient, this method is highly sensitive to noise and cannot detect slow-moving objects. More specifically, frame differencing only detects temporal variation between consecutive frames and fails when objects become static. Background subtraction instead models the scene structure and allows detection of both moving and newly introduced static objects. Moreover, selective background updating reduces ghost artifacts.

Background Subtraction. A more robust approach maintains an explicit background model $Bg(x, y)$ and compares each frame against it:

$$D_t(x, y) = |I_t(x, y) - Bg(x, y)| \tag{2}$$

The key challenge is building and maintaining an accurate background model that adapts to gradual changes (illumination variations, moved furniture) while remaining stable against temporary changes (moving persons, shadows).

2.1.2 Background Model Initialization and Update

Our approach uses a **median-based initialization** followed by **adaptive updating**:

Initialization. The background is estimated as the median of the first N frames:

$$Bg(x, y) = \text{median}\{I_1(x, y), I_2(x, y), \ldots, I_N(x, y)\} \tag{3}$$

The median operator is preferred over the mean because it is robust to outliers (moving objects during initialization). With $N = 100$ frames at 12 FPS, this corresponds to approximately 8 seconds of initialization period.

Adaptive Update. To accommodate gradual environmental changes, the background is updated using exponential averaging:

$$Bg_{t+1}(x, y) = \begin{cases} (1 - \alpha)Bg_t(x, y) + \alpha I_t(x, y) & \text{if } (x, y) \in \text{Static (static pixel)} \\ Bg_t(x, y) & \text{otherwise} \end{cases} \tag{4}$$

where $\alpha \in (0, 1)$ is the learning rate. A pixel is classified as static if:

- Intensity difference is below a low threshold: $|I_t(x, y) - Bg_t(x, y)| < \tau_{\text{low}}$
- Not part of a confirmed foreground object

The learning rate $\alpha = 0.08$ provides a time constant of approximately $1/\alpha \approx 12$ frames (1 second), allowing slow adaptation while maintaining stability. This selective update strategy prevents "ghost" effects where stationary foreground objects would eventually be absorbed into the background.

2.2 Edge Detection

Edges represent significant local changes in image intensity and are fundamental features for object detection and recognition. Edge detection aims to identify pixels where the intensity gradient magnitude is locally maximal.

2.2.1 Gradient-Based Edge Detection

The image gradient at pixel (x, y) is defined as:

$$\nabla I = \begin{bmatrix} \frac{\partial I}{\partial x} \\ \frac{\partial I}{\partial y} \end{bmatrix} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} \tag{5}$$

The gradient magnitude and direction are:

$$|\nabla I| = \sqrt{G_x^2 + G_y^2} \tag{6}$$

$$\theta = \arctan\left(\frac{G_y}{G_x}\right) \tag{7}$$

2.2.2 Sobel Operator

The Sobel operator approximates the gradient using 3×3 convolution kernels that incorporate Gaussian smoothing:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} * I, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} * I \tag{8}$$

The Sobel kernels can be decomposed as the product of a smoothing and a differentiation operation:

$$G_x = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & +1 \end{bmatrix}, \quad G_y = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ +1 \end{bmatrix} \tag{9}$$

Advantages of Sobel:

- **Integrated smoothing:** The averaging operation (weights 1-2-1) reduces noise sensitivity
- **Directional information:** Provides both horizontal (G_x) and vertical (G_y) gradients
- **Computational efficiency:** Fast convolution-based implementation
- **Established baseline:** Well-studied with known characteristics

Limitations:

- Produces thick edges (typically 3-5 pixels wide)
- No explicit edge thinning or localization
- Still sensitive to high-frequency noise despite smoothing

2.2.3 Dual Detection Strategy

For robust change detection, we combine intensity-based and edge-based differences:

$$D_{\text{intensity}}(x, y) = |I_t(x, y) - Bg(x, y)| \tag{10}$$

$$D_{\text{edge}}(x, y) = ||\nabla I_t(x, y)| - |\nabla Bg(x, y)|| \tag{11}$$

The final foreground mask combines both criteria:

$$F(x, y) = \begin{cases} 255 & \text{if } D_{\text{intensity}}(x, y) > \tau_d \text{ AND } D_{\text{edge}}(x, y) > \tau_e \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

This dual-threshold approach significantly reduces false positives: both intensity and structure must change simultaneously for a pixel to be classified as foreground. The edge difference is particularly robust to gradual illumination changes that affect intensity uniformly but preserve edge structure.

2.3 Mathematical Morphology

Mathematical morphology is a powerful framework for analyzing and processing binary images based on geometric structure and shape. Morphological operations probe an input image $A \subseteq E^2$ using a predefined structuring element (or kernel) B, which encodes spatial relationships.

2.3.1 Basic Operations

Erosion. The erosion of A by B is defined as:

$$A \ominus B = \{c \in E^2 : \ c + b \in A, \ \forall b \in B\} \tag{13}$$

Erosion shrinks objects by removing boundary pixels. Intuitively, the structuring element B, translated by c, must be entirely contained in A for c to belong to the eroded set. As a result, erosion eliminates small protrusions, removes thin connections, and separates weakly connected components. If the structuring element contains the origin, the eroded set is always a subset of the original one ($A \ominus B \subseteq A$), making erosion an anti-extensive operation.

Dilation. The dilation of A by B is defined as:

$$A \oplus B = \{c \in E^2 : \ c = a + b, \ a \in A, \ b \in B\} = \bigcup_{b \in B} A_b = \bigcup_{a \in A} B_a \tag{14}$$

Dilation expands objects by adding pixels to their boundaries. Operationally, it places the structuring element B at every foreground pixel of A and takes the union of all translated copies. Dilation is particularly useful for:

- Filling small holes and gaps,
- Connecting nearby components,
- Compensating for fragmentation introduced during binarization.

If the structuring element contains the origin, dilation is an extensive operation ($A \subseteq A \oplus B$). However, dilation is less selective than erosion: while it effectively reconnects fragmented regions, it may also distort object shapes by enlarging boundaries and over-smoothing details.
Project context. In this project, dilation is used after binarization and segmentation to improve the continuity of the extracted person mask. In particular, it helps refine the subject's

silhouette by filling small gaps along the body contour and strengthening weak edges caused by thresholding noise.

2.3.2 Compound Operations

Opening. Opening is defined as erosion followed by dilation:

$$A \circ B = (A \ominus B) \oplus B \tag{15}$$

Opening removes small objects and thin protrusions while preserving the overall size and shape of larger structures. Since it begins with erosion, it eliminates small foreground components that cannot fully contain the structuring element; the subsequent dilation partially restores the remaining objects.
Opening is anti-extensive ($A \circ B \subseteq A$) and acts as a shape-preserving smoothing filter. It breaks narrow connections, removes small isolated regions, and smooths object boundaries.
In practice, opening is often applied immediately after image binarization to remove noise-induced structures and improve segmentation quality.

Closing. Closing is defined as dilation followed by erosion:

$$A \bullet B = (A \oplus B) \ominus B \tag{16}$$

Closing fills small holes and gaps within objects, connects nearby components, and smooths internal boundaries. Since dilation is applied first, small background intrusions are filled; the subsequent erosion restores the object contour while preserving the filled regions.
Closing is an extensive operation ($A \subseteq A \bullet B$). It is particularly effective for repairing small background discontinuities inside foreground objects and for improving mask compactness without significantly altering global shape.

2.3.3 Structuring Element Selection

The choice of structuring element shape and size critically affects results:

- **Shape:**
  - *Rectangular:* Fastest computation, but introduces directional bias
  - *Elliptical:* Approximates circular, suitable for organic shapes (persons), no directional preference
  - *Custom:* Can be designed for specific applications
- **Size:** Trade-off between noise removal and detail preservation

- Small kernels (3×3 to 7×7): Preserve fine details, limited noise removal
- Medium kernels (9×9 to 15×15): Balanced approach for typical applications
- Large kernels (17×17+): Aggressive smoothing, may lose shape details

For person detection, elliptical kernels are preferred as they better match the rounded shapes of human bodies without introducing the anisotropy of rectangular kernels.

2.3.4 Area Opening

Area opening is a connected-component based operation that removes all connected components with area less than a threshold λ:

$$\gamma_\lambda(A) = \bigcup \{C \mid C \text{ is a connected component of } A, |C| \geq \lambda\} \tag{17}$$

Unlike standard opening with a fixed kernel, area opening is **shape-preserving**: it removes small objects entirely but does not alter the shape of objects larger than λ. This makes it particularly effective for noise removal in detection applications.

2.3.5 Multi-Pass Morphological Pipeline

Our system employs a carefully designed morphological pipeline:

1. **Area Opening (λ):** Remove small noise components
2. **Closing (B_1, n_1 iterations):** Fill holes, connect fragmented parts
3. **Opening (B_2, n_2 iterations):** Remove protrusions, smooth boundaries

The iteration count controls the strength of each operation: multiple iterations with a small kernel approximate a single iteration with a larger kernel but provide finer control over the progression.

2.4 Connected Component Labeling

Connected component labeling assigns unique identifiers to each maximal set of connected foreground pixels in a binary image. This enables extraction of individual objects for subsequent analysis.

2.4.1 Connectivity Definitions

Two pixels p and q are connected if:

- **4-connectivity:** They share an edge (horizontally or vertically adjacent)

$$N_4(p) = \{(x \pm 1, y), (x, y \pm 1)\} \tag{18}$$

- **8-connectivity:** They share an edge or corner (includes diagonals)

$$N_8(p) = N_4(p) \cup \{(x \pm 1, y \pm 1)\} \tag{19}$$

Our system uses 8-connectivity as it better captures the full extent of objects and avoids artificial fragmentation that can occur with 4-connectivity, especially for diagonal edges or rotated objects.

2.4.2 Two-Pass Labeling Algorithm

The classical two-pass algorithm operates as follows:

Algorithm 1 Two-Pass Connected Component Labeling

```
First Pass: Scan image left-to-right, top-to-bottom
for each foreground pixel p do
    Find labels of already-processed neighbors N(p)
    if no labeled neighbors then
        Assign new label L_next
    else
        Assign minimum neighbor label L_min
        Record equivalences between neighbor labels
    end if
end for
Equivalence Resolution: Build union-find structure
Second Pass: Replace provisional labels with final labels
```

Modern implementations (e.g., OpenCV's `connectedComponentsWithStats`) use optimized union-find structures achieving near-linear time complexity $O(n)$ where n is the number of pixels.

2.4.3 Component Statistics

For each connected component, features are computed from the refined contour C_i (obtained from the binary mask, and all the morphologic operations), using standard OpenCV geometric primitives.

Geometric Properties:

- **Area:** $A = \texttt{contourArea}(C_i)$
- **Bounding Box:** $(x, y, w, h) = \texttt{boundingRect}(C_i)$
- **Centroid (box-center):** $(c_x, c_y) = \left(x + \frac{w}{2},\, y + \frac{h}{2}\right)$
- **Perimeter:** $P = \texttt{arcLength}(C_i)$

Shape Descriptors:

- **Extent:** $E = \dfrac{A}{w \cdot h}$
- **Solidity:** $S = \dfrac{A}{A_{\text{hull}}}$, where A_{hull} is computed from the convex hull of C_i
- **Aspect Ratio:** $AR = \dfrac{h}{w}$

These descriptors support rule-based classification. In the proposed configuration, person candidates must satisfy a minimum size constraint ($A \geq 1200$ and $h \geq 80$) together with shape constraints ($S \geq 0.15$, $E \geq 0.10$, $1.0 \leq AR \leq 4.0$, and $h > 0.8w$). Additional early rejection rules discard components that are too small, too horizontal, or too fragmented.
which reflect the expected geometric structure of upright human silhouettes.

2.5 Summary

The theoretical concepts covered in this section form the foundation of our intrusion detection system:

- **Change detection** via background subtraction with adaptive updating
- **Edge detection** using Sobel operator for structural change analysis
- **Mathematical morphology** for noise reduction and shape regularization
- **Connected component labeling** for object extraction and feature computation

The next section describes how these techniques are integrated into a coherent detection pipeline optimized for the intrusion detection task.

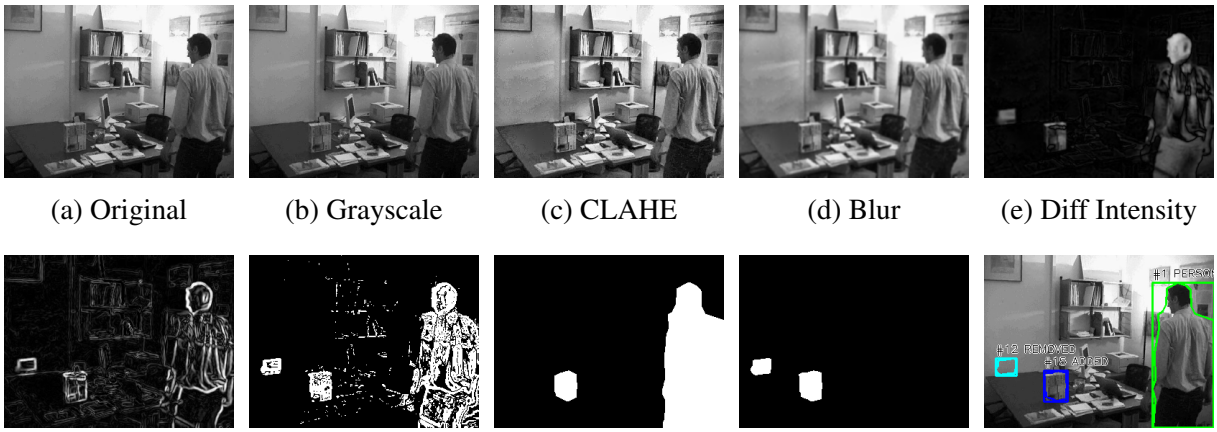
3 Methodology

This section presents the complete intrusion detection pipeline, translating the theoretical foundations introduced in Section 2 into a practical real-time system. Each stage of the processing workflow is described in detail, highlighting its role within the overall architecture and motivating the main design choices.

Both tasks considered in this work are addressed throughout the discussion. When necessary, task-specific differences are explicitly clarified, while the section titles indicate the particular task to which the description refers.

3.1 System Overview

The detection pipeline processes each video frame through ten sequential stages, visualized in Figure 1. These stages transform the raw RGB input into classified, tracked detections with temporal consistency.



(a) Original (b) Grayscale (c) CLAHE (d) Blur (e) Diff Intensity

(f) Diff Edge (g) Foreground Raw (h) FG Person (i) FG Objects (j) Final Result

Figure 1: Complete processing pipeline from raw input to final detection. Each stage progressively refines the segmentation: (a) Original frame (loaded as 3-channel BGR by OpenCV despite the video being 8-bit grayscale), (b) grayscale conversion to ensure single-channel format, (c) CLAHE contrast enhancement, (d) Gaussian smoothing, (e) intensity difference, (f) edge difference using Sobel, (g) combined foreground mask, (h) morphologically refined person mask, (i) object mask, (j) final tracked detections with labels.

3.2 Background Modeling

3.2.1 Initialization

Following the approach described in the course material on change detection, the background model is initialized using the median operator over $N = 100$ frames as reported in (3)

Rationale: The median is preferred over the mean because it is inherently robust to outliers, this idea comes form the *median filter* used for mitigating the noise in an image, in fact using a

mean filter the outcome will be influencing the pixel near to the outlier, concluding the person can be considered as an exogenous impulsive noise. During initialization, moving objects (persons entering/exiting) appear in some frames but not all. The median effectively discards these transient elements, producing a clean static background even when the scene is not entirely empty.
With the video captured at 12 FPS, 100 frames correspond to approximately 8.3 seconds—sufficient time to capture lighting variations while short enough for practical , moreover an almost full empty background can be appreciated only when the person exit from the FOV of the camera.

3.2.2 Adaptive Update

To accommodate gradual environmental changes—such as slowly varying illumination due to sunlight movement or minor scene modifications (e.g., furniture repositioning)—the background model is updated using exponential moving average, as commonly employed in motion detection systems as reported in (4)
Learning Rate Selection: The time constant for adaptation is $\tau = 1/\alpha \approx 12$ frames (1 second at 12 FPS). This value balances:

- **Adaptation speed:** Sufficiently fast to track gradual illumination changes (e.g., clouds passing over windows, natural light decay)
- **Stability:** Sufficiently slow to prevent foreground absorption. If α were too large (e.g., 0.2), stationary persons would contaminate the background within seconds, causing *ghosting*

Selective Update Strategy: The update is applied only to "static" pixels, defined as pixels where:

1. Intensity difference is below a low threshold: $|I_t(x, y) - Bg_t(x, y)| < \tau_{\text{low}}$ where $\tau_{\text{low}} = 4$
2. The pixel does not belong to a confirmed foreground object

This selective mechanism prevents active foreground regions from corrupting the background model, even if they remain stationary for extended periods.
Low Threshold Selection ($\tau_{\text{low}} = 4$): The selective update threshold is set to 4 gray levels, corresponding to approximately twice the standard deviation of typical sensor noise ($\sigma_{\text{noise}} \approx 2$ gray levels). This value ensures that:

- Intensity variations due to sensor noise ($< 2\sigma$) are absorbed into the background, maintaining model stability
- True intensity changes ($\geq 2\sigma$) prevent background update, preserving foreground integrity

The threshold is fixed rather than adaptive because sensor noise characteristics are relatively uniform across the indoor scene with stable camera settings. Adaptive thresholding based on local variance estimation would increase computational cost without significant benefit in this controlled environment.

3.3 Preprocessing

3.3.1 Grayscale Conversion

RGB frames are converted to grayscale using standard luminance weighting:

$$I_{\text{gray}} = 0.299R + 0.587G + 0.114B \tag{20}$$

The formula (20) respect the standard ITU-R BT.601.

This reduces dimensionality while preserving perceptual brightness.

Note on Input Format: Although the provided video is recorded as 8-bit grayscale (256 intensity levels) as specified in the project documentation, OpenCV's VideoCapture interface loads all video frames in BGR format with three channels by default. The grayscale conversion step (Eq. 22) ensures the image is in single-channel format, which is required for subsequent preprocessing (CLAHE) and edge detection (Sobel) operations.

3.3.2 CLAHE (Contrast Limited Adaptive Histogram Equalization)

Indoor environments often exhibit non-uniform illumination with dark corners and bright windows. Standard histogram equalization would over-amplify noise in dark regions. CLAHE addresses this by:

- Dividing the image into 8×8 tiles,
- Applying histogram equalization locally to each tile,
- Limiting contrast amplification with clip_limit $= 2.0$ (it is a OpenCV default parameter),
- Interpolating tile boundaries for smooth transitions.

Figure 2 illustrates the improvement in local contrast.

(a) Before CLAHE (b) After CLAHE

Figure 2: Effect of CLAHE preprocessing. (a) Original grayscale with poor contrast in darker regions. (b) After CLAHE: enhanced local contrast while avoiding noise amplification through clip limiting.

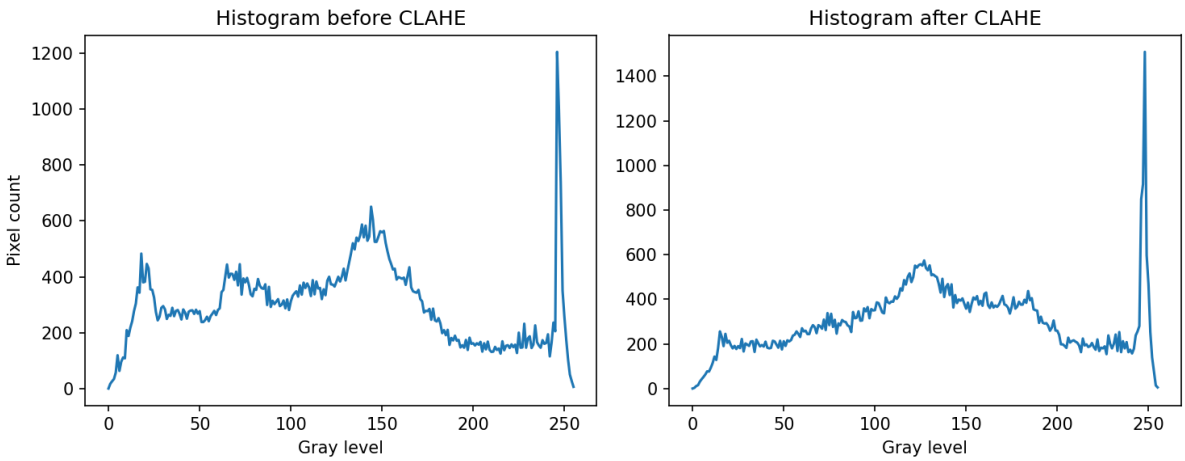


Figure 3: CLAHE before and after gray level istogram (frame 380)

CLAHE operates locally by equalizing small image tiles independently. This enhances low-contrast regions by redistributing intensity values within each tile, while the clip limit prevents noise amplification, it is crucial for indoor surveillance where lighting conditions vary across the scene. Although the global histogram remains largely similar, local contrast improvement leads to stronger gradient responses and more robust edge-based change detection

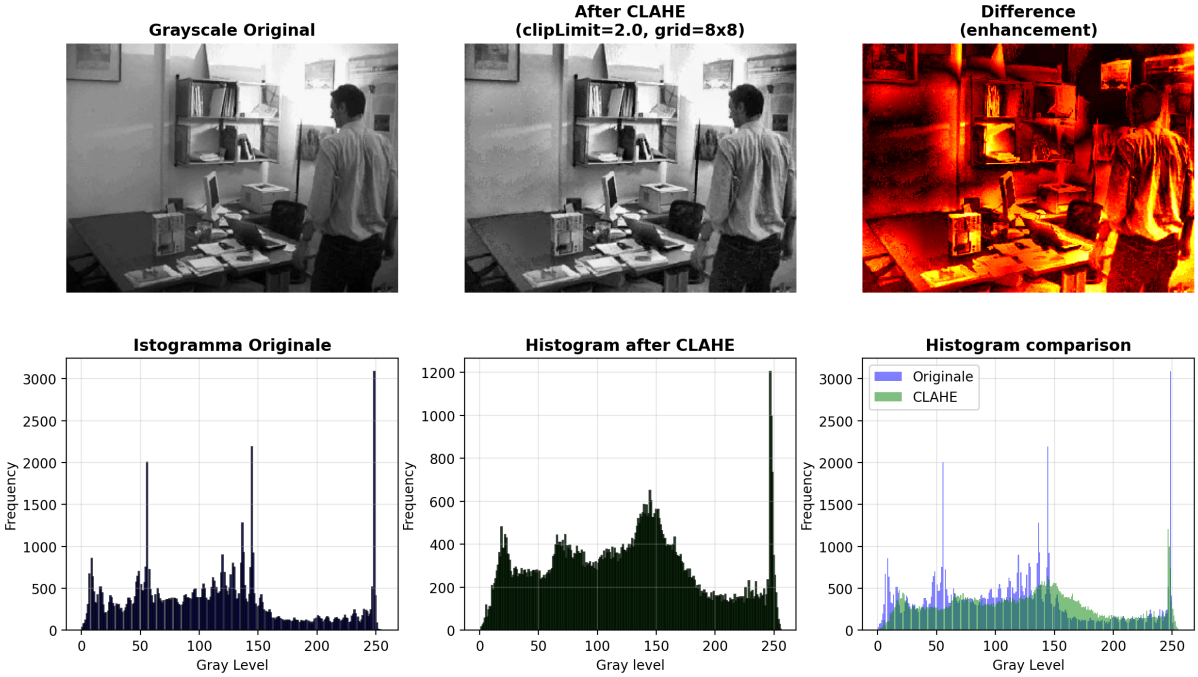


Figure 4: CLAHE effect on local contrast enhancement. (a,b) Before/after comparison illustrating enhanced visibility in low-contrast regions. CLAHE operates locally, applying limited adjustments in uniformly dark areas while performing stronger intensity redistribution in brighter regions to improve contrast without amplifying noise. (c,d) Histograms demonstrating local redistribution while preserving global distribution.

3.3.3 Gaussian Smoothing

A 5×5 Gaussian filter is applied to reduce high-frequency noise prior to edge detection:

$$I_{\text{blur}} = G_{5\times5} * I_{\text{CLAHE}} \tag{21}$$

The continuous two-dimensional Gaussian kernel is defined as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \tag{22}$$

In digital image processing, a discrete approximation of this kernel is sampled over a finite window of size $k \times k$, producing a normalized convolution mask whose coefficients sum to one. The kernel size represents a trade-off between noise suppression and detail preservation. Smaller kernels retain fine structural details but are less effective at removing noise, whereas larger kernels provide stronger smoothing at the cost of potential edge attenuation. The selected 5×5 kernel offers a balanced compromise between these two effects.

The Gaussian filter is preferred over a simple mean (box) filter due to its superior frequency response. Specifically, the Gaussian transfer function decays monotonically with increasing frequency, resulting in smooth attenuation of high-frequency components. In contrast, the mean filter exhibits oscillatory behavior in the frequency domain, introducing secondary lobes that

may generate artificial details or ringing artifacts not present in the original image.
As the standard deviation σ increases, image structures are progressively smoothed but not geometrically distorted, since the Gaussian kernel performs weighted averaging with radially symmetric weights. Larger kernels reduce truncation effects at the filter boundaries but increase computational cost. However, this cost can be mitigated thanks to the separability property of the Gaussian kernel:

$$G(x, y) = G_x(x) \cdot G_y(y) \tag{23}$$

which allows the 2D convolution to be implemented as two successive 1D convolutions.
In the implementation, the value of σ is automatically determined by the OpenCV function `GaussianBlur` based on the specified kernel size. For a 5×5 kernel, this corresponds approximately to $\sigma \approx 1.1$.
Conceptually, Gaussian smoothing performs a weighted sum of pixel intensities within a local neighborhood, assigning larger weights to pixels closer to the center and progressively smaller weights to those farther away.
Gaussian vs Bilateral filtering: While bilateral filtering offers edge-preserving smoothing, it is not employed here because: (1) edge preservation is unnecessary before edge detection—the Sobel operator explicitly computes edges from the smoothed image, rendering pre-preserved edges redundant; (2) bilateral filtering is computationally expensive (5-10× slower than Gaussian) and would hinder real-time processing (12 FPS requirement); (3) isotropic smoothing is actually desirable to uniformly attenuate high-frequency noise before gradient computation. The separable Gaussian filter provides an optimal trade-off between noise suppression and computational efficiency.

3.4 Dual Change Detection

Following the course material on background subtraction, we implement a **dual detection strategy** combining intensity and structural information.

3.4.1 Intensity Difference and Edge Difference via Sobel Operator

The intensity difference (Figure 1(e)) captures regions where pixel values have changed, equation (10) compute this difference
This is effective for detecting uniform objects but sensitive to illumination changes.
As presented in the course, the Sobel operator computes image gradients using separable 3×3 kernels (equarion (8)). The edge magnitude difference (Figure 1(f)) is reported in equation (11):
Why Sobel? Among gradient operators (Roberts, Prewitt, Sobel, Laplacian), Sobel offers:

- Integrated smoothing (1-2-1 weighting) reducing noise sensitivity
- Directional information (G_x, G_y) useful for orientation analysis

- Computational efficiency ($O(n)$ convolution)
- Well-established performance characteristics

Figure 5 shows that Sobel effectively captures person contours even in low-contrast regions where intensity difference alone fails.

(a) Intensity Only (b) Edge (Sobel) (c) Combined (AND)

Figure 5: Dual detection strategy. (a) Intensity difference captures overall shape but misses low-contrast edges. (b) Sobel edge difference preserves structural information even with varying illumination. (c) Combined result (logical AND) reduces false positives while maintaining detection sensitivity.

Why not sharpening? Edge enhancement filters (e.g., unsharp masking) amplify high-frequency components indiscriminately, including sensor noise present in low-light indoor video. Since our objective is robust structural change detection rather than visual enhancement, we rely on the Sobel operator, which inherently balances edge detection with noise suppression through its integrated smoothing kernel (weights 1-2-1). This approach avoids noise amplification while maintaining adequate edge sensitivity.

3.4.2 Dual Thresholding

This step constitutes the binarization of the change maps into a binary foreground mask.
The final foreground mask requires both criteria to be satisfied according to (12) with $\tau_{\text{int}} = 18$ and $\tau_{\text{edge}} = 12$.
Motivation: The AND logic significantly reduces false positives from:

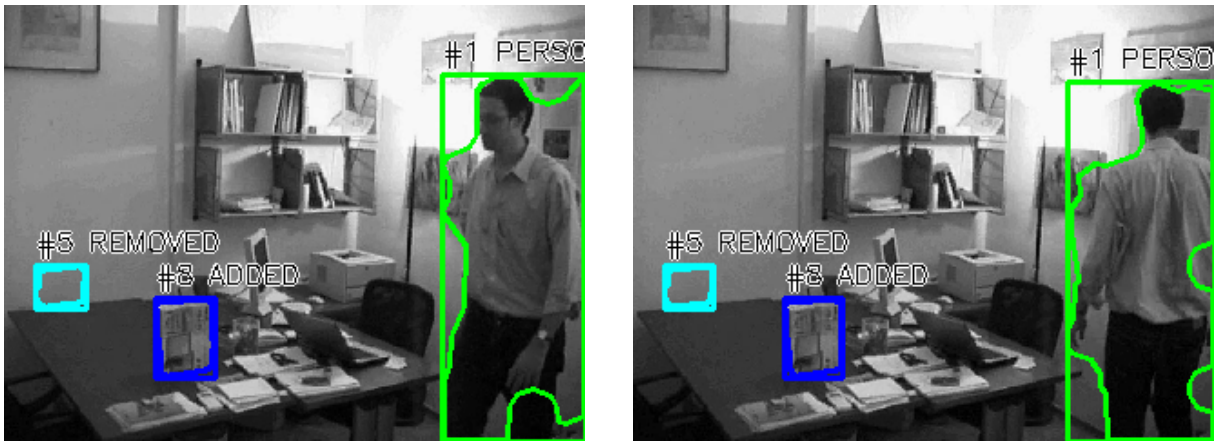
- Shadows (change intensity but not edges)
- Gradual illumination changes (preserve edge structure)
- Camera noise (random intensity fluctuations without coherent structure)

3.4.3 Threshold Selection Methodology

The threshold values $\tau_{\text{int}} = 18$ and $\tau_{\text{edge}} = 12$ were determined through systematic analysis combining histogram inspection and empirical validation. This subsection provides detailed

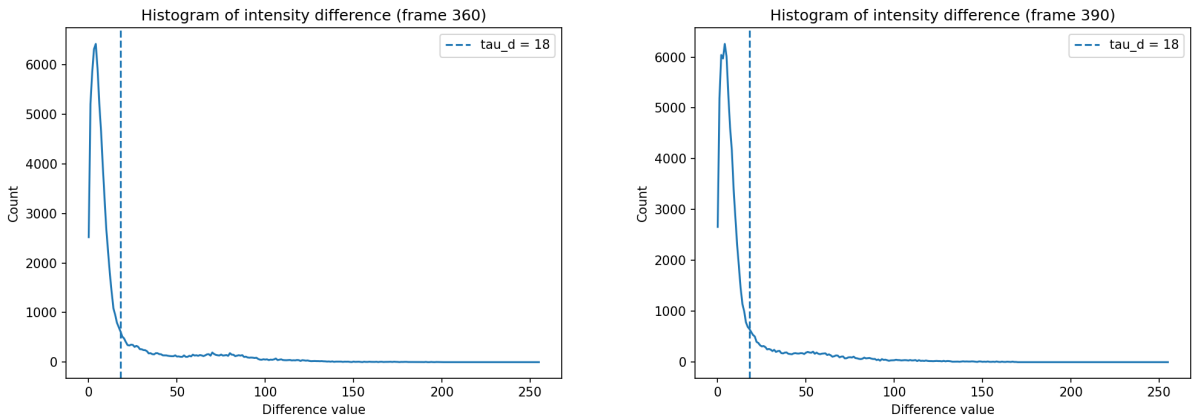
justification for these choices and explains why fixed thresholds are preferred over adaptive methods for this application.

Histogram Analysis of Difference Images Figure 7 shows the histogram distribution of intensity difference values $D_{\text{int}}(x, y) = |I_t(x, y) - B(x, y)|$ for representative frames 360 and 390, when the person is present in the scene.



(a) Frame 360 (b) Frame 390

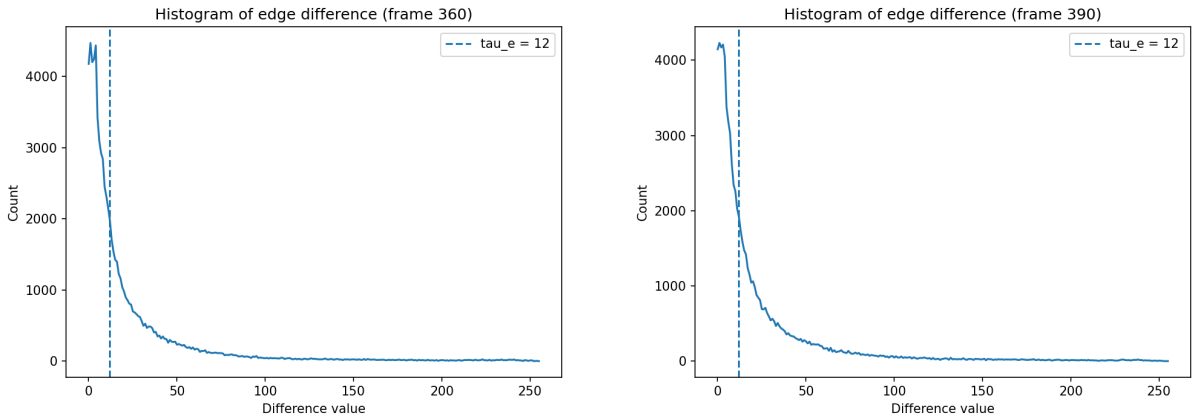
Figure 6: Result at frames 360 and 390



(a) Frame 360 (b) Frame 390

Figure 7: Histogram of intensity difference D_{int} showing exponential decay from background pixels (near zero) to foreground pixels (higher values). The vertical dashed line indicates $\tau_{\text{int}} = 18$. **X-axis:** Absolute intensity difference value in gray levels [0-255]. **Y-axis:** Number of pixels (count) exhibiting each difference value.

Similarly, Figure 8 presents the histogram distribution of edge difference values $D_{\text{edge}}(x, y) = ||I_t(x, y)| - |B(x, y)||$.



(a) Frame 360

(b) Frame 390

Figure 8: Histogram of edge difference D_{edge} computed using Sobel operator. The vertical dashed line indicates $\tau_{\text{edge}} = 12$. **X-axis:** Absolute edge magnitude difference [0-255]. **Y-axis:** Number of pixels with each edge difference value.

Interpretation of Histogram Distributions **Why a single peak instead of bimodal distribution?**

Classical thresholding theory (as presented in Chapter 6 of the course material on image segmentation) assumes bimodal histograms with two distinct peaks:

- First peak: Background pixels (low difference values)
- Valley: Optimal threshold location
- Second peak: Foreground pixels (high difference values)

However, our histograms exhibit a **single dominant peak near zero followed by exponential decay**, rather than two distinct peaks. This distribution is characteristic of **sparse foreground scenarios** and can be explained as follows:

1. **Dominant background class:** The vast majority of pixels (>84%) belong to the static background, producing the strong peak at $D_{\text{int}} \approx 0$. With 320×240 = 76,800 total pixels and the person occupying approximately 11,000-12,000 pixels, background pixels outnumber foreground pixels
2. **Gradual foreground transition:** Foreground pixels do not form a compact intensity cluster. Instead, they exhibit a wide range of difference values due to:
    - Varying clothing/skin tones (different local contrasts with background)
    - Lighting gradients across the person's body
    - Partial occlusions and depth variations
    - Motion blur at edges

This dispersion prevents formation of a second distinct peak, resulting in the observed exponential tail.

3. **Edge effects:** Even within the person's silhouette, internal regions with low texture (e.g., uniform clothing) produce small differences, while boundaries produce large differences. This heterogeneity spreads foreground pixels across the histogram rather than concentrating them.

Empirical Threshold Positioning Given the exponential decay distribution, the threshold is positioned at the **transition point** where the background noise distribution ends and the foreground signal begins. Specifically:

- **Intensity difference ($\tau_{\text{int}} = 18$):** The histogram shows rapid decay from the peak at zero, reaching a plateau (low but constant count) around $D_{\text{int}} \approx 15 - 20$. This plateau represents true foreground pixels. Setting $\tau_{\text{int}} = 18$ ensures:

$$\tau_{\text{int}} \approx 2\sigma_{\text{background}} + \text{margin} \tag{24}$$

  where $\sigma_{\text{background}}$ is the standard deviation of background intensity variations due to lighting non-uniformity and sensor noise (typically $\sigma \approx 5 - 8$ gray levels).

- **Edge difference ($\tau_{\text{edge}} = 12$):** Edge magnitudes are generally smaller than intensity differences (Sobel produces normalized gradients). The threshold $\tau_{\text{edge}} = 12$ is positioned where the exponential tail stabilizes, capturing significant structural changes while rejecting weak edges from texture and noise.

Results: The combination $(\tau_{\text{int}} = 18, \tau_{\text{edge}} = 12)$ achieved optimal balance:

- Lower thresholds ($\tau_{\text{int}} < 16$): Excessive noise (false positives on textured background)
- Higher thresholds ($\tau_{\text{int}} > 20$): Missing body parts (especially low-contrast clothing)
- Edge threshold $\tau_{\text{edge}} = 12$: Captures structural boundaries without being oversensitive to weak edges

Rationale for Fixed Thresholds Unlike adaptive methods (e.g., Otsu thresholding which dynamically computes thresholds from histogram modes), fixed thresholds were selected for this application due to:

1. **Stable indoor environment:** The scene exhibits relatively consistent illumination. While gradual brightness changes occur (darkening over time), these are handled by:
   - CLAHE preprocessing (local contrast normalization)

- Adaptive background update (Equation 4)
- Dual detection AND logic (illumination changes affect intensity but preserve edge structure)

2. **Sparse foreground limitation:** Adaptive thresholding methods like Otsu assume comparable class sizes to identify the inter-class variance maximum. With foreground occupying only $\sim$15% of pixels, Otsu would likely fail to separate classes correctly, potentially positioning the threshold too low (capturing noise) or too high (missing foreground).

3. **Deterministic behavior:** Fixed thresholds ensure reproducible results across runs. Adaptive thresholds can fluctuate frame-to-frame based on scene content (e.g., large foreground objects temporarily shift histogram), leading to temporal instability in detection.

4. **Dual detection compensates:** The AND logic between intensity and edge criteria provides inherent robustness:

   - If illumination brightens gradually, D_{int} may increase across the entire scene, but D_{edge} remains stable (edge structure preserved)
   - Only regions with *both* intensity and structural changes trigger detection
   - This implicit adaptation reduces the need for dynamic threshold adjustment

3.5 Morphological Filtering

The raw foreground mask (Figure 1(g)) contains noise and fragmentation. Following the mathematical morphology theory presented in the course, we apply a multi-pass refinement pipeline.

3.5.1 Two-Pass Strategy

To optimize parameters for different object types, we process persons and generic objects separately:

Algorithm 2 Two-Pass Morphological Pipeline

Pass 1: Person Detection

$F_{person} \leftarrow \text{AreaOpening}(F, \lambda = 500)$ $F_{person} \leftarrow \text{Closing}(F_{person}, K^{9\times9}_{close}, n = 5)$ $F_{person} \leftarrow \text{Opening}(F_{person}, K^{5\times5}_{open}, n = 4)$ $M_{person} \leftarrow \text{Dilate}(F_{person}, K^{19\times19}_{ellipse}, n = 1)$ *// spatial safety margin before object subtraction*

Pass 2: Object Detection

$F_{objects} \leftarrow F \wedge \neg M_{person}$ *// subtract dilated person mask* $F_{objects} \leftarrow \text{AreaOpening}(F_{objects}, \lambda = 200)$ $F_{objects} \leftarrow \text{Closing}(F_{objects}, K^{11\times11}_{close}, n = 3)$ $F_{objects} \leftarrow \text{Opening}(F_{objects}, K^{5\times5}_{open}, n = 1)$

F_{person} and F_{object} are just binary mask obtained formally

- F_{person}, is a binary mask containing pixels identified as persons (value 1 for person, 0 otherwise). It results from: Dual detection $\rightarrow$ Morphology $\rightarrow$ Area opening $\rightarrow$ Classification,

- F_{person} is a binary mask containing blobs classified as "other", computed as: $F_{\text{objects}} \leftarrow F \wedge \neg F_{\text{person}}$ (total foreground minus person mask).

3.5.2 Parameter Selection Rationale

Area Opening: Unlike standard morphological opening, area opening (Equation 17) is shape-preserving: it removes components with area $< \lambda$ entirely but does not alter the shape of larger components. This makes it ideal for noise removal without distorting object boundaries.
Kernel Sizes:

- **Elliptical shape:** Matches organic human body contours better than rectangular kernels, avoiding directional bias

- **9×9 for closing:** Large enough to bridge typical gaps in person silhouette caused by low contrast clothing

- **5×5 for opening:** Removes noise protrusions while preserving anatomical details (arms, legs)

Iteration Counts: Multiple iterations with medium kernels provide better control than single iteration with very large kernels. The effective kernel size is approximately: $K_{\text{eff}} \approx K \times n$.
Critical Balance: Initial experiments with more aggressive parameters (15×15, 4 iterations) successfully removed noise but caused over-smoothing, resulting in loss of anatomical details. The current parameters (9×9, 4-5 iterations) provide optimal trade-off: adequate noise suppression while maintaining shape fidelity.

(a) Before Morphology (b) After Person Morphology (c) After Object Morphology

Figure 9: Morphological refinement progression. (a) Raw foreground: noisy but preserves structure. (b) Person mask after closing+opening: holes filled, shape consolidated, main contours preserved. (c) Object mask: lighter morphology preserves smaller details.

3.6 Classification and Feature Extraction

3.6.1 Connected Component Labeling

After morphological refinement, the binary foreground mask is processed using 8-connectivity labeling as described in Section 2.4.1.

Let C_i denote the i-th maximal 8-connected component extracted from the refined foreground mask. Each C_i represents a candidate object (person or other item) prior to semantic classification.

The implementation relies on contour extraction via OpenCV, which internally follows the same theoretical principles as the classical two-pass connected component labeling algorithm (Section 2.4). Each contour corresponds to one connected component C_i.

3.6.2 Feature Computation

For each connected component C_i, features are computed from the refined contour using standard OpenCV geometric primitives.

Table 1: Extracted features and their discriminative purpose

Feature	**Definition**	**Purpose**
Area	$A = \texttt{contourArea}(C_i)$	Size filtering
Bounding Box	$(x, y, w, h) = \texttt{boundingRect}(C_i)$	Spatial extent
Aspect Ratio	$AR = h/w$	Vertical vs horizontal shape
Extent	$E = A/(w \cdot h)$	Bounding box occupancy
Solidity	$S = A/A_{\text{hull}}$	Shape compactness
Perimeter	$P = \texttt{arcLength}(C_i)$	Boundary smoothness

The centroid used for tracking is approximated as the center of the bounding box:

$$(c_x, c_y) = \left(x + \frac{w}{2},\ y + \frac{h}{2}\right)$$

This approximation is computationally efficient and sufficiently accurate for tracking purposes.

3.6.3 Person Classification Rules

Classification is rule-based and directly follows the implementation in the code.

The decision logic includes both early rejection rules and shape validation:

Algorithm 3 Person Classification (Implementation-Aligned)

```
if A ≥ 3000 and h ≥ 80 and ω ≥ 50 and AR ≥ 1.0 then
    return "person"
end if
if A < 1200 or h < 50 then
    return "other"
end if
if ω > h then
    return "other"
end if
if S < 0.12 or E < 0.08 then
    return "other"
end if
bigEnough ← (A ≥ 1200 and h ≥ 80)
goodShape ← (S ≥ 0.15 and E ≥ 0.10 and 1.0 ≤ AR ≤ 4.0 and h > 0.8w)
if bigEnough and goodShape then
    return "person"
else
    return "other"
end if
```

Rationale:

- $A \geq 1200$: Ensures sufficient spatial support for human silhouettes at typical indoor distances.
- $h \geq 80$: Enforces minimum vertical extent.
- $w \leq h$: Rejects horizontally elongated objects.
- $S \geq 0.15$ and $E \geq 0.10$: Ensure compact and coherent shapes.
- $1.0 \leq AR \leq 4.0$: Encodes expected human vertical elongation.
- $h > 0.8w$: Reinforces vertical dominance.

Objects failing any of these constraints are labeled as "other".

Additional rejection rules for generic objects. After person classification, each candidate blob in $F_{objects}$ is further filtered by a dedicated rejection function that discards:

- **Undersized blobs:** width or height < 10 px.

- **Border blobs:** bounding box within 12 px of the image boundary, which typically correspond to partial occlusions at the frame edge rather than actual objects.

- **Incoherent shapes:** solidity < 0.25 or extent < 0.15, thresholds stricter than those for the person class, since generic objects (boxes, bags) are expected to be more compact.

This rejection stage operates independently from person classification and applies exclusively to the object pipeline (Algotithm 2).

This category includes static objects (added items), removed-object candidates (handled separately), and residual noise components.

The classification employs a two-tier strategy:

- **Tier 1 (Priority):** Large objects ($A \geq 3000$ pixels) are immediately classified as persons without shape validation, reducing false negatives from unusual poses

- **Tier 2 (Standard):** Smaller objects require strict shape constraints to avoid misclassifying compact items

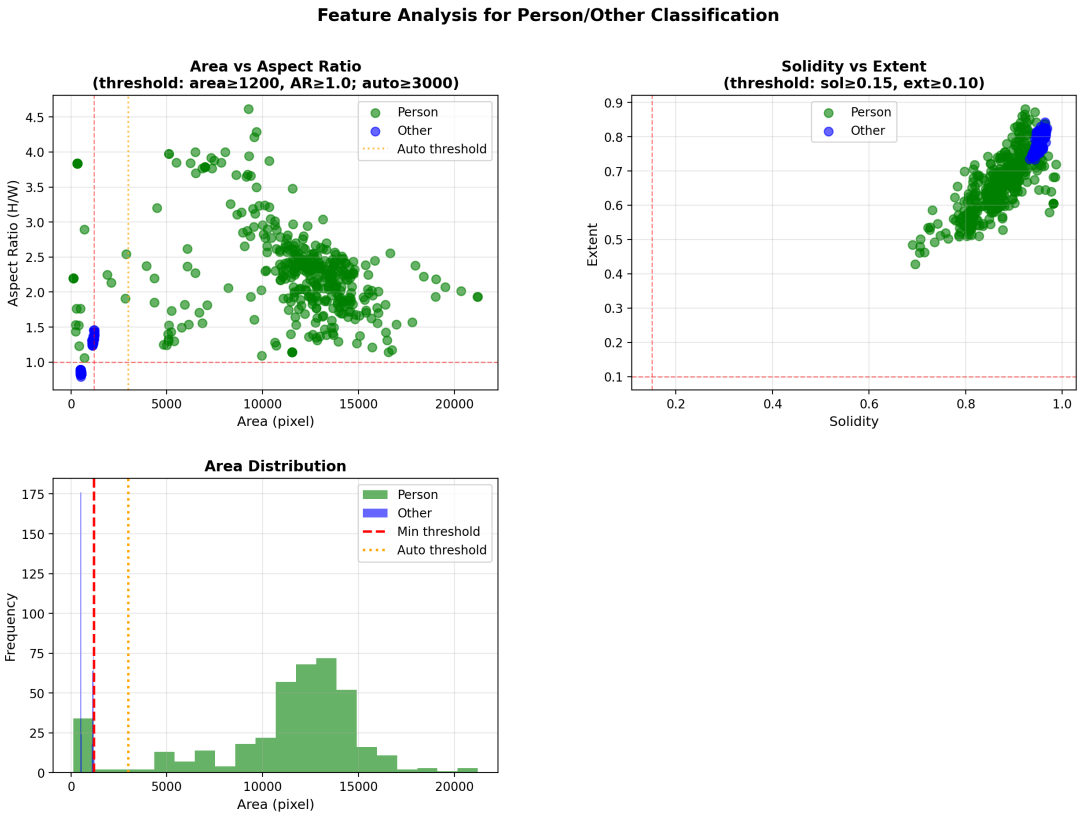


Figure 10: Feature space analysis for person/object classification. (a) Area vs. Aspect Ratio scatter plot showing a clear separation between persons (green, vertically elongated) and other objects (blue, smaller and more compact). Red dashed lines indicate the classification thresholds (Area $\geq$ 1200, AR $\geq$ 1.0). The vertical red line represents the minimum area required to classify a detection as a person, while the horizontal red line reflects the aspect ratio criterion, distinguishing vertically elongated bounding boxes (typical of persons) from more horizontally distributed ones (typical of objects). The orange line represents the automatically optimized classification boundary used for person identification. (b) Solidity vs. Extent plot illustrating shape compactness differences: persons exhibit lower solidity (0.6–0.85) due to their concave body contours, whereas objects are generally more convex (0.8–1.0), typically having rectangular or square shapes. An extent value close to 1 indicates that the bounding box is almost completely filled; lower values indicate empty regions within the bounding box. (c) Area distribution histogram confirming that the threshold at 1200 pixels effectively separates the two classes with minimal overlap. The textbox summarizes the complete rule-based classification criteria implemented in the system.

Although the separation is clear, some detections classified as “Person” appear on the left-hand side of the Area vs. AR plot (below the area threshold). This occurs because the person is also detected while exiting the scene, as shown in frame 250.

Figure 11: Person identification while exiting the scene

From Figure 11, it can be observed that the person's bounding box has a size comparable to the background object (the one being removed from the table). Some of these borderline classifications may also be caused by the crouching movement performed while reaching under the table, which reduces the bounding box area and alters its aspect ratio.
In Figure 10(b), objects typically have square or rectangular shapes and therefore tend to fill the bounding box almost completely. In contrast, the human silhouette is generally concave (due to arms, legs, and body shape), producing empty regions between the smooth contour and the bounding box. Only when the person is close to the camera does the bounding box become more uniformly filled.
From Figure 10(c), the threshold at 1200 pixels:

- captures approximately 95% of the objects (below the threshold),
- captures approximately 98% of the persons (above the threshold).

The thresholds were determined empirically by repeatedly running simulations and tuning the corresponding parameters to achieve optimal separation.

Quantitative Metrics from the Feature Space Analysis

- Total detections: 775 (403 Person + 372 Other)
- Person detections: 403 (52%)
- Other detections: 372 (48%)

The dataset is therefore well balanced (52/48 split), indicating that the system is not biased toward either class.

Feature Space Separation Summary

1. **Area vs. Aspect Ratio:** Near-linear separation using thresholds (1200, 1.0). Estimated overlap: $\approx$ 3–5% (few person samples below threshold).

2. **Solidity vs. Extent:** Good separation, though with larger overlap. Person cluster: (0.6–0.85, 0.5–0.85) Other cluster: (0.8–1.0, 0.8–0.95) Estimated overlap: $\approx$ 10–15%.

3. **Area Distribution:** Excellent separation. Overlap below 5% in the 1000–2000 pixel range.

3.7 Temporal Tracking

To maintain persistent IDs across frames and handle occlusions, we implement IoU-based tracking.
Each detected object is assigned a unique track ID when first appearing in the scene. This ID is maintained across frames as long as the object remains trackable, enabling:

- **Temporal consistency:** The same object retains the same ID across the video

- **Multi-object distinction:** Multiple objects in the same frame have different IDs

- **Re-identification:** Briefly occluded objects recover their original ID upon reappearance

The track ID is incremented globally: the first detected object receives ID=1, the second ID=2, and so on. Track IDs are displayed in the output video as labels (e.g., "#1 PERSON", "#3 ADDED", "#5 REMOVED").
ID lifecycle:

1. **Creation:** New detection with no matching existing track $\rightarrow$ assign new ID

2. **Update:** Detection matches existing track via IoU $\rightarrow$ keep same ID, update position

3. **Missed:** Track not matched for 1-20 frames $\rightarrow$ ID persists, counter incremented

4. **Deletion:** Track missed for > 20 frames (1.67s) $\rightarrow$ ID permanently deleted

5. **Re-detection:** Previously deleted object reappears $\rightarrow$ receives new ID (not recovered)

3.7.1 IoU-Based Association

Temporal association between detections and existing tracks is performed using the **Intersection-over-Union (IoU)** metric, a standard similarity measure for comparing spatial regions in object detection and tracking.
Given two bounding boxes B_1 and B_2, IoU is defined as:

$$\text{IoU}(B_1, B_2) = \frac{|B_1 \cap B_2|}{|B_1 \cup B_2|} \tag{25}$$

where $|B_1 \cap B_2|$ denotes the area of intersection between the two bounding boxes and $|B_1 \cup B_2|$ denotes the area of their union. Each bounding box is parameterized as $B_i = (x, y, w, h)$, representing top-left coordinates, width, and height.
Interpretation. IoU measures the degree of spatial overlap between two regions and produces a normalized score in the range $[0, 1]$:

- IoU $= 0 \rightarrow$ no overlap
- IoU $= 1 \rightarrow$ perfect alignment

Unlike raw distance metrics (e.g., centroid displacement), IoU accounts for both position and spatial extent, making it particularly suitable for bounding box matching.
For each detection d_t at frame t, IoU is computed against all active tracks T_k. The detection is associated with track T_k if:

$$k^* = \arg\max_k \text{IoU}(d_t, T_k)$$

subject to the constraint:

$$\text{IoU}(d_t, T_{k^*}) > \tau_{\text{IoU}}$$

with $\tau_{\text{IoU}} = 0.2$.
The relatively low threshold is motivated by the low frame rate (12 FPS), which can produce noticeable inter-frame displacement. A permissive threshold allows robust association even under rapid motion or partial overlap reduction.
Why IoU? IoU provides robustness to:

- **Scale changes:** If the person moves toward or away from the camera, bounding boxes change size but still maintain significant overlap.
- **Shape deformation:** Pose changes alter the bounding box geometry but preserve spatial consistency.
- **Partial occlusions:** Even when part of the object is hidden, remaining overlap often exceeds the threshold.

Compared to centroid-based matching, IoU better captures spatial consistency in non-rigid objects such as persons.

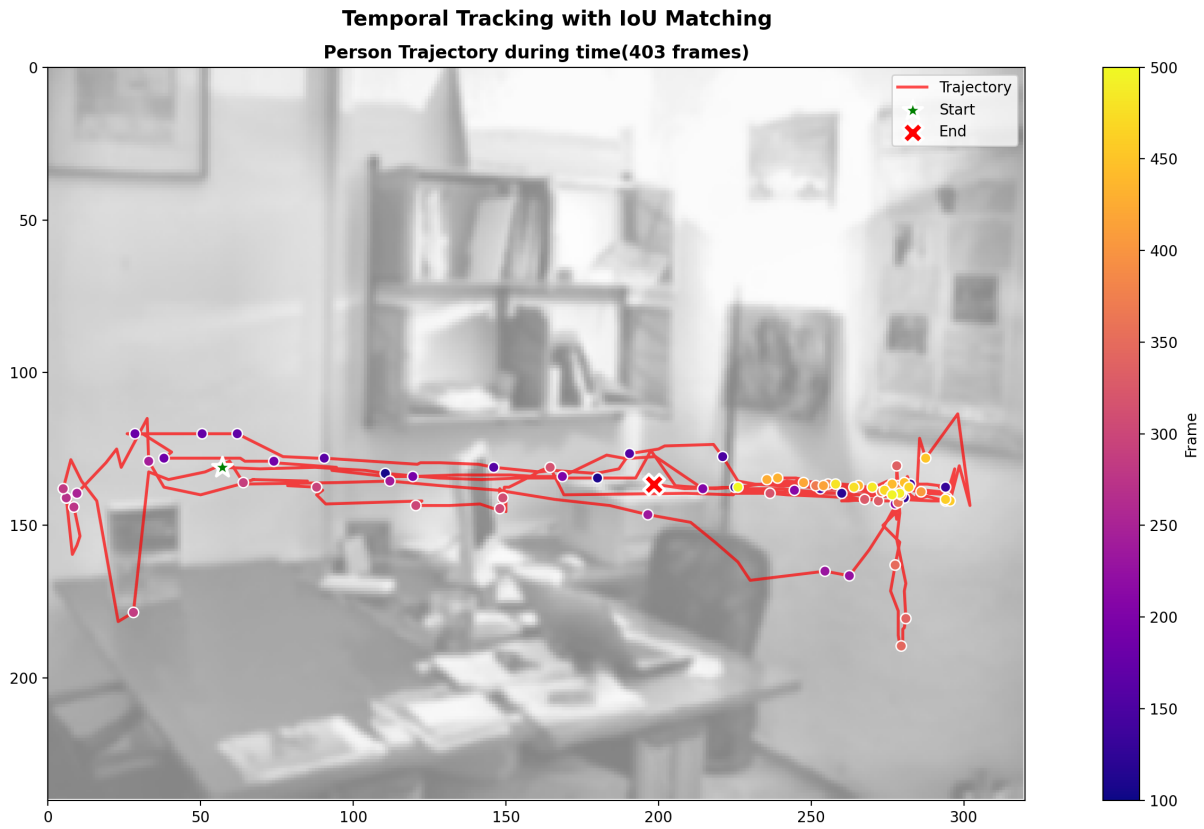


Figure 12: Temporal tracking behavior of the person during the video

Static Person Suppression

To avoid false persistent detections, a person track is removed from the output visualization if its centroid displacement has remained below 3 px for 20 or more consecutive frames. Formally, let $\Delta_t = \sqrt{(\Delta c_x)^2 + (\Delta c_y)^2}$ be the centroid displacement at frame t. A counter n_{static} is incremented whenever $\Delta_t < 3$, and reset to zero otherwise:

$$n_{static}^{t+1} = \begin{cases} n_{static}^{t} + 1 & \text{if } \Delta_t < 3\,\text{px} \\ 0 & \text{otherwise} \end{cases} \tag{26}$$

The track is suppressed from rendering (but not deleted) when $n_{static} \geq 20$. This prevents a stationary person—already absorbed into the background model via selective update—from generating a spurious persistent bounding box.

3.7.2 Occlusion Handling

Each active track maintains a "missed" counter incremented when no detection is associated in the current frame. Tracks are deleted only after 20 consecutive missed frames (≈ 1.6 seconds

at 12 FPS).
This temporal tolerance allows recovery from short-term occlusions (e.g., person partially hidden behind furniture) while preventing indefinite persistence of stale tracks.
This temporal filtering eliminates transient noise while maintaining low latency for genuine intrusions.

3.8 Removed Object Detection

In the second part of the sequence, the subject removes an object that was originally part of the background. In this scenario, classical foreground detection identifies a blob corresponding to the newly revealed background region. However, this blob does not represent a newly introduced object, but rather the disappearance of a previously static one.
From a modeling perspective, removal detection is complementary to the standard background subtraction formulation (2). While foreground detection relies on the absolute residual $|I_t - B_t|$, removal detection exploits the **signed residual**:

$$D_{\text{removed}}(x, y) = B_t(x, y) - I_t(x, y) \tag{27}$$

A positive value of $D_{\text{removed}}(x, y)$ indicates that the current frame is darker than the background model at that location. This condition typically occurs when an object previously present in the background is removed, revealing an underlying darker surface.
In the implementation, the signed residual is:

- thresholded to retain only significant positive differences,
- converted into a binary mask,
- processed independently from the standard foreground mask.

Thus, D_{removed} denotes the **binary mask of removal candidates**, i.e., pixels potentially corresponding to object disappearance.
This explicit separation ensures that removal detection does not interfere with person or added-object tracking.

3.8.1 Person Exclusion

A critical issue arises from the motion of the person. Shadows, pose variations, or transient occlusions may generate positive signed residuals, leading to false removal detections.
To suppress these artifacts, an exclusion mask is constructed from the refined person mask F_{person}:

Algorithm 4 Person Exclusion

1: $M_{\text{person}} \leftarrow \text{Dilate}(F_{\text{person}}, K_{45\times45}, n = 3)$
2: $D_{\text{removed}} \leftarrow D_{\text{removed}} \wedge \neg M_{\text{person}}$

Here:

- F_{person} is the binary mask of the detected person,
- M_{person} is its dilated version,
- $\neg M_{\text{person}}$ denotes the logical complement of the dilated mask.

The dilation (45×45 kernel, three iterations) intentionally enlarges the person region, creating a spatial safety margin around the silhouette. This prevents removal candidates located near the moving person from being considered valid.
The logical operation

$$D_{\text{removed}} \leftarrow D_{\text{removed}} \wedge \neg M_{\text{person}}$$

is applied pixel-wise and retains removal candidates only in regions spatially disjoint from the (expanded) person area.
This strategy ensures that removal detection operates exclusively on static portions of the scene, reducing false positives caused by motion artifacts or shadow effects.

3.8.2 Dedicated Morphological Processing

Removed objects are typically smaller than the person and may appear fragmented after thresholding. Therefore, the implementation applies a morphology pipeline distinct from the one used for person detection:

- **Closing (9×9 rectangular kernel, n=3):** fills internal holes in thin objects.
- **Opening (7×7 elliptical kernel, n=2):** removes isolated shadow artifacts.
- **Area filtering:** connected components with area < 150 px are discarded as noise; components with area > 2000 px are also rejected, as blobs of that size are more likely caused by the person's shadow or large illumination artifacts than by a genuine removed object.

This pipeline mirrors the strategy described in Section 2.3 but with parameters adapted to small compact blobs instead of vertically elongated human silhouettes.
After refinement, connected component labeling is applied, and removed objects are inserted into a dedicated tracking list. In the output video, they are rendered in cyan to visually distinguish them from persons (green) and added objects (blue).

3.8.3 Integration with Selective Background Update

Removal detection is tightly coupled with the selective background updating mechanism defined in Eq. (4). The goal is to decide, for each pixel (x, y), whether the background model should be updated or frozen in the current frame.
To this end, a binary update mask $M_{\text{update}}(x, y) \in \{0, 1\}$ is constructed:

$$M_{\text{update}}(x, y) = \begin{cases} 1 & \text{pixel allowed to update} \\ 0 & \text{pixel frozen} \end{cases}$$

The construction follows a priority-based logic implemented in the code:

Algorithm 5 Selective Background Update Mask Construction

1: $M_{\text{update}} \leftarrow \mathbf{1}$ {Initialize: all pixels updatable}
2: $M_{\text{update}}[F_{\text{person}} \cup F_{\text{objects}}] \leftarrow 0$
3: $M_{\text{update}}[|I_t - Bg_t| < 4] \leftarrow 1$
4: **for** each confirmed track T_k **do**
5: $\quad M_{\text{update}}[\text{bbox}_{T_k} \pm 8\text{px}] \leftarrow 0$
6: **end for**

Interpretation.

- **Foreground freezing:** Pixels belonging to detected persons or objects are excluded from update to prevent foreground absorption (ghosting).
- **Noise compensation:** Pixels with small intensity variation ($|I_t - Bg_t| < 4$) are re-enabled for update, allowing the model to absorb sensor noise and maintain stability.
- **Track protection:** Regions corresponding to confirmed tracks are further protected with spatial padding to avoid accidental background contamination due to temporary detection loss.

Finally, the background is updated pixel-wise:

$$Bg_{t+1}(x, y) = M_{\text{update}}(x, y) \cdot ((1 - \alpha)Bg_t(x, y) + \alpha I_t(x, y)) + (1 - M_{\text{update}}(x, y))Bg_t(x, y)$$

with learning rate $\alpha = 0.08$, corresponding to a time constant $\tau \approx 12$ frames.

Role in Removal Detection. After a removed object is confirmed (12 consecutive frames), its region becomes eligible for update again. Since the object is no longer present, the exponential update gradually reconstructs the clean background within approximately one second.

This controlled interaction between removal detection and selective background updating allows the system to distinguish between:

- true object removal,
- temporary occlusions,
- illumination fluctuations.

Temporal behavior. The interaction between removal detection and background updating follows a priority-based logic:

- **Added objects:** Once confirmed (15 consecutive frames), their regions remain permanently excluded from background updating, preventing ghost absorption.
- **Removed objects:** After confirmation (12 consecutive frames), the corresponding pixels are allowed to update again. Since the object is no longer present, the exponential update gradually reconstructs the clean background within approximately one second.
- **Noise handling:** Pixels satisfying $|I_t - Bg_t| < 4$ ($\approx 2\sigma_{noise}$) are always updated to stabilize the model.
- **Transient artifacts:** Tentative tracks are excluded from update but automatically deleted after 20 missed frames, preventing long-term corruption of the background model.

This tight coupling ensures that the system reliably distinguishes true removed objects (persistent disappearance) from temporary occlusions and illumination fluctuations. In the final visualization, each removed object is assigned a persistent track ID, consistent with the IoU-based association described in Section 3.7.

Figure 13: Final results

4 Conclusions

4.1 Summary of Achievements

This project presents an intrusion detection system for indoor surveillance based on classical computer vision techniques. All mandatory requirements have been satisfied, and the optional removed-object detection task has been successfully implemented.
The main contributions can be summarized as follows:

1. **Dual change detection:** The combination of intensity and edge-based differences improves robustness to illumination changes and noise while preserving structural consistency.

2. **Morphological refinement:** A carefully tuned two-stage morphological pipeline effectively removes noise and preserves meaningful object shapes.

3. **Rule-based classification:** Geometric and shape descriptors (area, aspect ratio, solidity, extent) enable reliable separation between persons and other objects.

4. **Temporal tracking:** IoU-based association ensures temporal consistency and stable object identities across frames.

5. **Removed object detection:** Inverse differencing with person exclusion allows identification of removed objects without introducing significant false positives.

4.2 Limitations

The system is designed for a controlled indoor environment and relies on fixed thresholds and parameters. Changes in camera position, resolution, or lighting conditions may require parameter adjustment.
The rule-based classification approach, while effective for the given scenario, lacks the adaptability of learning-based methods. Additionally, prolonged occlusions result in track deletion and ID reassignment.

4.3 Final Remarks

The results confirm that classical computer vision techniques remain effective for structured surveillance scenarios when properly designed and tuned. The integration of background subtraction, edge detection, morphology, and temporal tracking provides a consistent and interpretable solution suitable for real-time applications.